Light Collection in a Polished Plastic Scintillator Read Out by Side-Mounted SiPMs: a Geant4 Matrix Simulation Study

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Abstract

We describe a Geant4 simulation of a 250 mm×250 mm×10 mm polished plastic scintillator (PVT, UPS-92S/EJ-200 class) read out by side-coupled silicon photomultipliers (SiPMs), modelled on the muon-trigger detector of Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)]. The detector is exposed to a vertical 4 GeV muon pencil beam. We present the geometry and the optical-physics implementation in detail—the Geant4 *unified* surface model, the finish/type combinations used to represent four wrapping configurations (bare/black holder, TiO_2 paint, aluminium foil, glossy paper), the photocathode efficiency model that drives photodetection, and the per-event optical accounting that feeds the analysis. A 48-file matrix scan (4 SiPM layouts × 4 reflectors × 3 beam positions, 1000 events each) yields collection efficiencies that reproduce the experimentally observed ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy, with the absolute scale set by the SiPM photocathode. We report per-event photon yields, photoelectron spectra, wavelength distributions, and position-resolved efficiency maps. Two implementation caveats are documented: a rare intermittent multi-threaded stall in the optical tracking, and a reversed centre/edge/corner trend in the TiO_2 configuration.

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1 Introduction and motivation

Plastic scintillators read out at their edges by SiPMs are attractive for large-area, low-cost muon triggers and veto systems. The amount of scintillation light that reaches the photosensor is governed almost entirely by how the non-instrumented faces of the plate are treated: a polished plate in air traps light by total internal reflection (TIR) and can be a nearly lossless light guide, whereas a plate wrapped in an absorbing or diffusive material loses photons at every bounce. Understanding this trade-off quantitatively is the central design question, and it is precisely where simulation is most useful, because the photon transport is a multi-bounce random walk that is difficult to model analytically.

This report documents a Geant4 (v11.3.2) simulation built to address that question for the specific geometry and surface treatments of Rao *et al.* [1]. The deliverable is a self-contained "matrix scan": every combination of four SiPM layouts and four reflector treatments, at three beam positions on the plate. The goal is not to fit a single number but to verify that the simulation reproduces the *ordering* of light-collection efficiency that the experiment observes, and to expose the physics behind it.

The remainder of this document is organised as follows. Section 2 describes the detector model. Section 3 details the optical physics implementation, verified line-by-line against the Geant4 source. Section 4 covers the simulation workflow and the matrix scan. Section 5 presents the results. Section 6 presents the whole-slab full-plate map. Section 7 documents known caveats.

2 Detector model

2.1 Scintillator plate

The active volume is a rectangular PVT scintillator $250 \times 250 \times 10\,\text{mm}$, mirroring the UPS-92S plate of Ref. [1]. It is assigned the Geant4 NIST base material `G4_PLASTIC_SC_VINYLTOLUENE` with a custom optical properties table [3, 4]:

- refractive index $n = 1.58$ (flat across the visible);
- light yield $1.0 \times 10^4\,\text{MeV}^{-1}$, single fast component, decay time 2.1 ns, rise time 0.9 ns;
- emission spectrum peaked at 425 nm (2.917 eV), essentially zero below 380 nm and above 550 nm;
- bulk attenuation length peaked at 3.8 m and falling steeply in the UV to model wavelength-shifter self-absorption;
- Birks constant $0.126\,\text{mm}\,\text{MeV}^{-1}$ (standard PVT value [5]).

Both scintillation and Čerenkov photons are produced and tracked (Sec. 3). The optical grid spans 310 nm to 700 nm so that the Čerenkov continuum, which extends well below the scintillation band, is covered.

2.2 SiPM readout and coupling

Each SiPM is modelled mechanically as a $4 \times 4 \times 0.6\,\text{mm}$ box of "SiPM epoxy" ($n = 1.55$), coupled to a side face of the plate through a 0.1 mm silicone grease pad ($n = 1.46$). The coupling chain is therefore *scintillator* $\rightarrow$ *grease* $\rightarrow$ *SiPM*, with Fresnel transmission at each dielectric interface and photodetection at the final grease/SiPM boundary.

Four layouts are studied (Figs. 14–17 and the schematic in Fig. 1):

Config 1 (8 SiPMs) two SiPMs per side face, at ±50% of the half-length.

Config 2 (4 SiPMs) one SiPM per side face, at the *same* ±50% offsets as config 1—i.e. config 1 with one device removed per side, the removed slot rotated from side to side so the layout remains four-fold symmetric.

Config 3 (2 SiPMs) both SiPMs on the *same* face—the top ($+y$) edge—at ±50%, matching T1/T2 of Ref. [1].

Config 4 (1 SiPM + WLS clamp) a single SiPM reading out a wavelength-shifting (WLS) fibre clamp along the $-x$ narrow face (Sec. 2.4).

SiPM footprints are clamped so they cannot overhang the slab face.

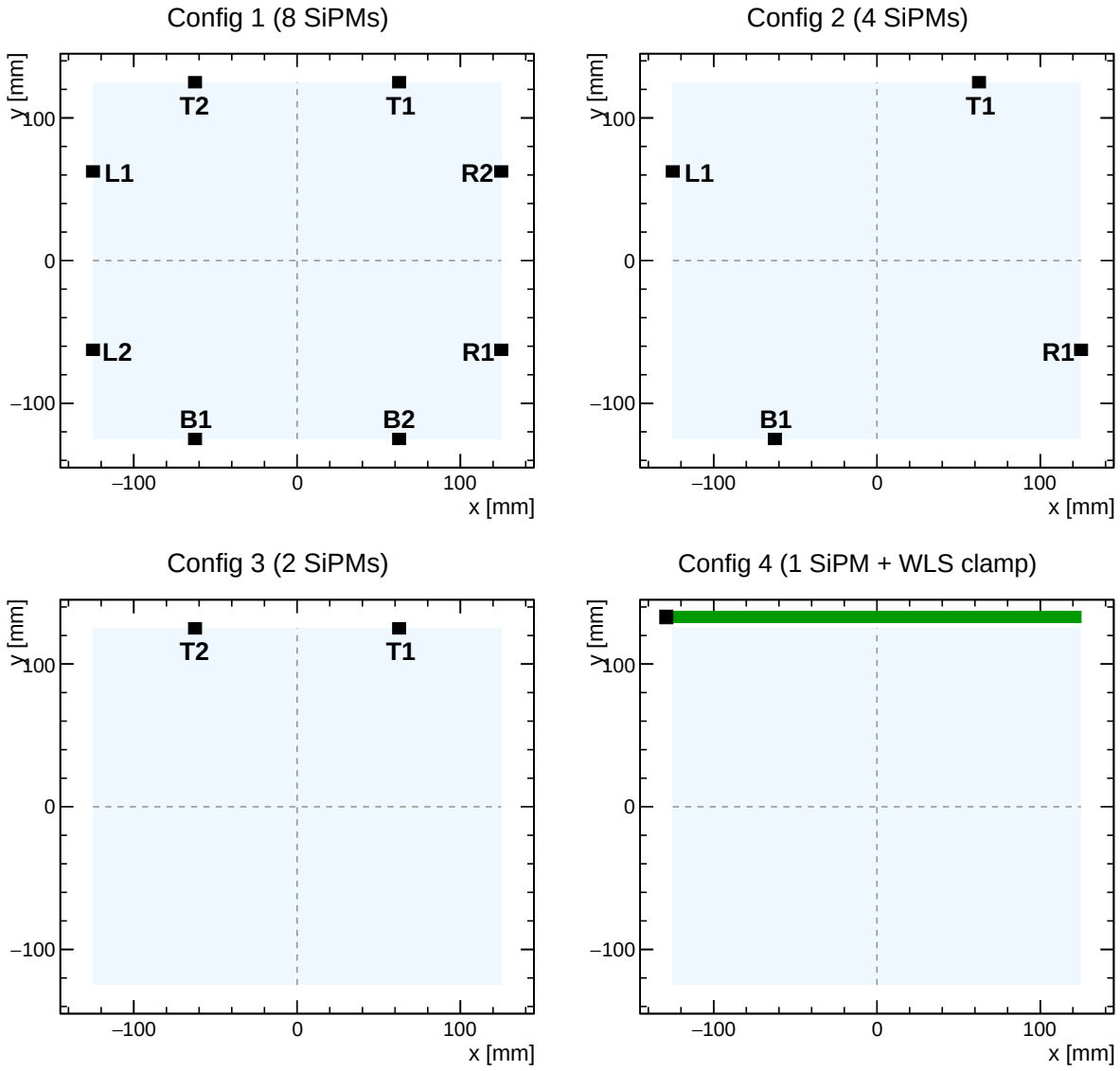


Figure 1: Schematic plan view of the four detector layouts on the 250 mm×250 mm plate footprint. Black squares mark the side-mounted SiPMs (drawn oversized for visibility; the active area is 4 mm×4 mm), and the dashed grey lines are the $x = 0$, $y = 0$ axes. Config 1 places two SiPMs on each of the four edges; config 2 places one SiPM per edge at the same offsets as config 1 (one device removed per side); config 3 places both SiPMs on the top ($+y$) edge; config 4 uses a single SiPM reading out a WLS fibre clamp, drawn along the top ($+y$) edge with the black read-out SiPM at its left end (green bar), following Fig. 3(b) of Ref. [1]. The inwards-offset labels name the eight side-mounted sensors (T/B/R/L = top/bottom/right/left); on each edge "1" marks the sensor nearer the corner reached when travelling clockwise around the perimeter, so "1" sits at the top-right (T1), bottom-right (R1), bottom-left (B1) and top-left (L1) corners. This is the scheme used in the light-sharing discussion below. The same black-square markers are overlaid on the collection efficiency maps (Figs. 14–17).

2.3 Reflector treatments

Four treatments are applied as a *logical skin* over the non-instrumented surfaces of the plate; the grease/SiPM patches override the skin locally with dedicated *border* surfaces. The mapping from the physical treatment to the Geant4 optical-surface parameters is the heart of the model and is summarised in Table 1 and justified in Sec. 3.

Table 1: Reflector optical-surface model. All surfaces use the *unified* model. $R(\lambda)$ is the tabulated reflectivity over 360 nm to 650 nm.

Config	Geant4 type	Geant4 finish	Key parameters
Bare / black holder	`dielectric_dielectric`	`polished` (absorber)	$R = T = 0$ (perfect sink)
TiO_2 paint	`dielectric_dielectric`	`groundfrontpainted`	Lambertian, $R \approx 0.955$–0.983
Aluminium foil	`dielectric_dielectric`	`polishedbackpainted`	specular spike = 1, $R \approx 0.84$–0.91
Glossy paper	`dielectric_dielectric`	`polishedbackpainted`	specular spike = 1, $R \approx 0.90$–0.955

A subtle but essential point concerns the *bare* configuration. A free-standing polished plate is an excellent TIR light guide and would be the *best* collector—contradicting the measurement, in which the bare plate gives the weakest signal. The resolution (Sec. 3) is that the bare plate is clamped into a black 3D-printed holder whose refractive index ($n \approx 1.55$) nearly matches the scintillator's. The faces are then in *effective optical contact* rather than in contact with air: TIR is *frustrated*, and light leaks out of the plate into the black walls, where it is absorbed. This is modelled by a perfect-absorber skin on the plate ($R = T = 0$) for the bare case.

2.4 Config 4: wavelength-shifting fibre clamp

The fourth layout follows the single-SiPM, WLS-readout variant of Ref. [1]. Four 1.0 mm-diameter Kuraray Y-11(200) wavelength-shifting fibres run the full 250 mm length of the $-x$ narrow face in a single row, touching one another (centre-to-centre pitch = 1.0 mm), so the bundle spans ≈ 4 mm of the 10 mm plate thickness. They are fully immersed in a 1.25 mm-thick grease layer between the plate and a black clamp bar outboard of it; the fibres are daughters of the grease volume, so the shared boundary avoids overlap and leaves a thin grease film on every side of each fibre (no clad↔clamp contact). The side of the clamp facing away from the plate carries a reflective skin, while the plate-facing side is open to the scintillator. Because the fibre cladding ($n = 1.49$) is slightly higher than the grease ($n = 1.46$), light trapped in a fibre core is still guided by total internal reflection. The single 4 mm×4 mm SiPM views all four fibre end faces at once at the $+y$ end; their far ($-y$) ends are closed by an aluminium mirror. The WLS process is enabled automatically when this geometry is built. Figure 2 shows an example event in this layout.

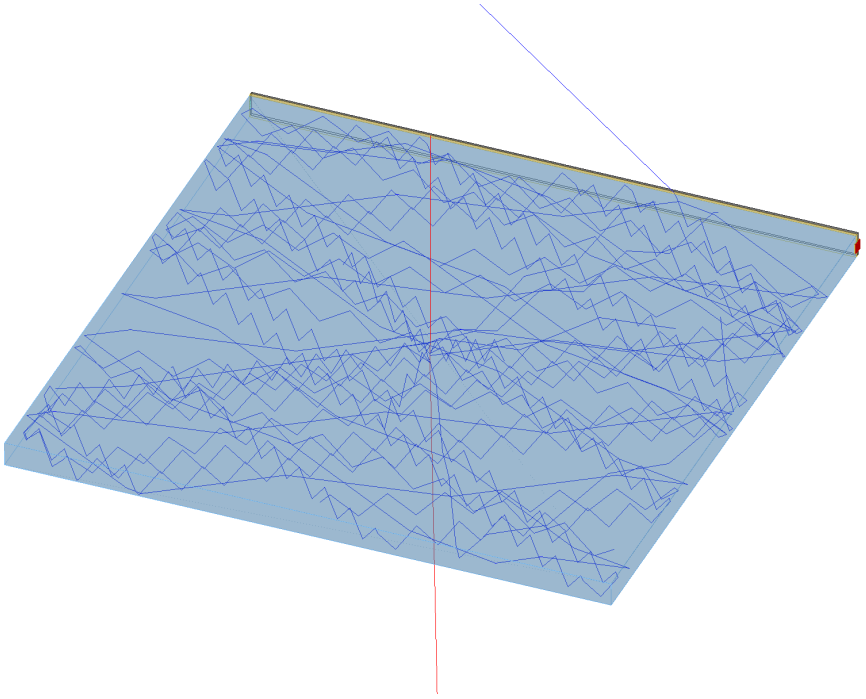
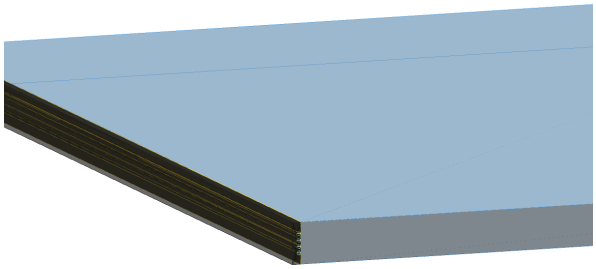
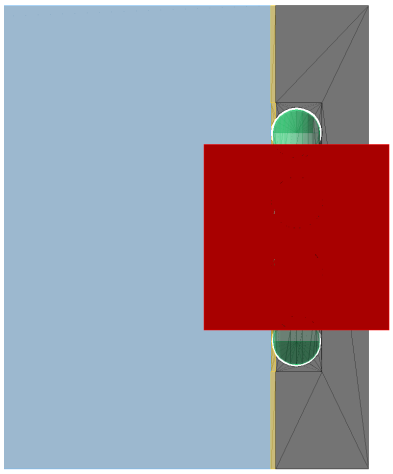


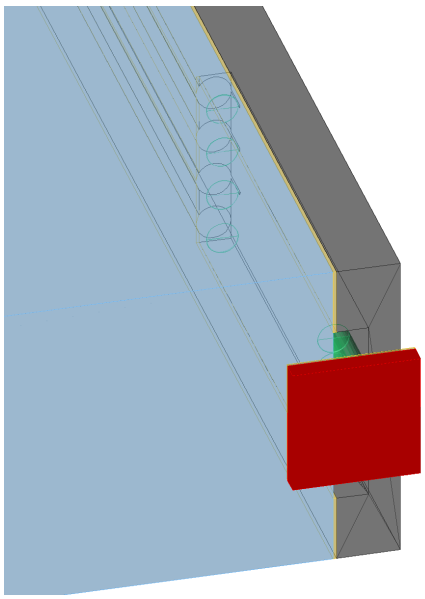
Figure 2: Head-less event display of a single muon in the config-4 WLS-fibre layout (same visualisation settings as Fig. 4). Scintillation photons enter the grease-filled clamp and are wavelength-shifted in the Y-11 fibres, whose re-emission is guided to the read-out SiPM.



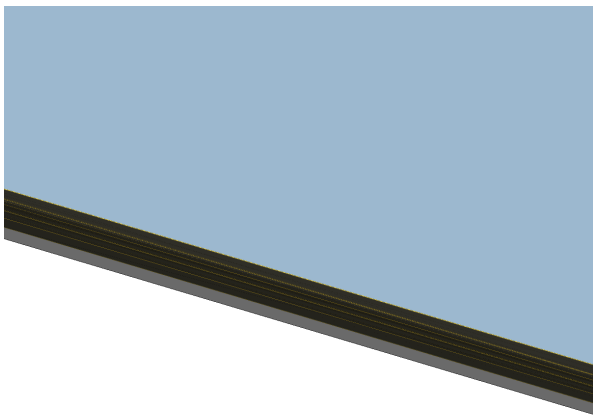
(a) Overview: the plate with the clamp bar running along its $-x$ face and the read-out sensor at the $+y$ end.



(b) Cross-section through the clamp: scintillator (light blue), the grease film (amber) that fills the slab–clamp gap, the four fibre discs (green) and the outboard black clamp. The fibres are fully immersed in the grease (no air gap, no clad–clamp contact).



(c) Read-out end at $+y$: the fibre bundle is coupled onto the single SiPM (red).



(d) Oblique view of the $-x$ face: the four fibres run the full 250 mm plate length in y.

Figure 3: Geometry-only close-ups of the config-4 WLS-fibre clamp (*no* beam-on). Fibre cores are pale green, cladding dark green, the grease film amber, the black clamp and holder dark grey, and the read-out SiPM red. The four touching 1.0 mm fibres sit on the $-x$ narrow face, embedded in the grease that completely fills the slab–clamp gap, read out at $+y$ by one SiPM and mirrored at $-y$.

3 Optical physics implementation

Optical photons are transported by the standard Geant4 optical physics module [2], with scintillation, Čerenkov, bulk absorption, Rayleigh scattering and boundary processes all active. Multiple scattering (Mie) is disabled; wavelength shifting (the `G4OpWLS` process) is enabled only for the config-4 fibre layout (Sec. 2.4) and inert for all others. This section describes the boundary physics and the photodetection model, and states explicitly the Geant4 behaviour that each modelling

choice relies on, verified against the Geant4 source in `.refs/source/processes/optical`.

3.1 The boundary process and the unified surface model

Photon crossing of a dielectric interface is handled by `G4OpBoundaryProcess`, which selects an `G4OpticalSurface` from (*i*) a `G4LogicalBorderSurface` between two physical volumes or (*ii*) a `G4LogicalSkinSurface` on a logical volume. Reading the source (lines 319–347), the lookup is: a border surface between the pre- and post-step volumes is consulted first; if none exists, the skin surface of the daughter (or, failing that, the mother) logical volume is used. This establishes two facts on which the geometry relies:

1. a `G4LogicalBorderSurface` takes precedence over a `G4LogicalSkinSurface` at the same boundary—hence the grease and SiPM patches punch "holes" in the wrapping skin;
2. if no surface object exists at a boundary, the process falls back to the default lossless TIR behaviour (via `DielectricDielectric()`).

The boundary process requires a valid `RINDEX` on both media; if the property is absent it sets status `NoRINDEX` and kills the photon (lines 289–306, 436–453). Every material in the model therefore carries an explicit `RINDEX` table.

For the *unified* model the probability weights for the reflection sub-models are read from the surface MPT—`SPECULARSPIKECONSTANT` (f_{ss}), `SPECULARLOBECONSTANT` (f_{sl}) and `BACKSCATTERCONSTANT` (f_{bs})—together with `REFLECTIVITY` (line 381) and, where applicable, `TRANSMITTANCE` (line 394) and `EFFICIENCY` (line 390). The surviving probability $1 - f_{ss} - f_{sl} - f_{bs}$ is shared between diffuse (Lambertian) and bare reflection.

3.2 Reflector finishes

The four treatments use three distinct code paths in `G4OpBoundaryProcess`, which we verified:

- `groundfrontpainted` (TiO_2). The finish is handled by the "painted-front" branch (lines 462–488): a uniform random draw against `REFLECTIVITY` decides absorbed / transmitted / reflected, and reflection is set to a pure Lambertian distribution (`LambertianReflection`, line 481). There is *no* TIR at this finish—photons are either diffusely reflected or absorbed. This is the physically correct description of sub-surface scattering in a thick TiO_2 paint.
- `polishedbackpainted` (Al, glossy paper). This finish is routed to `DielectricDielectric()` (line 456–459). The scintillator/air-gap interface still exhibits Fresnel reflection and TIR; sub-critical photons refract into the gap and are returned by the backing "paint" with reflectivity $R(\lambda)$ and a reflection distribution set by f_{ss}, f_{sl}, f_{bs}. We choose $f_{ss} = 1$ (pure specular spike) for aluminium, since a smooth metal foil is a mirror, and likewise for glossy paper—the experimental finding that Al $\approx$ glossy and both are the brightest means glossy paper is *not* behaving as a diffuser, so it is modelled as a slightly lossier mirror than Al.
- **Bare absorber**. This is deliberately *not* a wrapping skin. Instead a skin with $R = T = 0$ on the plate (and a matching skin on the holder shell) models the frustrated-TIR loss into the black holder.

3.3 Wavelength shifting

In the config-4 layout, photons that reach the fibre cores are absorbed and re-emitted isotropically at longer wavelength by the `G4OpWLS` process, which is configured through the fibre material's `WLSABSLENGTH`, `WLSCOMPONENT` and `WLSTIMECONSTANT` properties (Y-11 absorption peaks near 430 nm, re-emission near 490 nm). The process is switched on by the physics list when the WLS geometry is detected (the Geant4 optical-physics constructor reads the process activation at build time, so the flag is set before the base construction). Because the shifted emission no longer overlaps the fibre absorption band, the re-emitted light is transported losslessly along the fibre by TIR to the read-out SiPM—which is why the detected config-4 spectrum is green-shifted (mean $\sim$496 nm) relative to the direct-coupled configurations.

3.4 Photodetection: the SiPM photocathode

Photodetection is delegated entirely to Geant4, rather than being drawn manually in the sensitive detector. The grease/SiPM boundary is a `dielectric_metal` surface whose MPT carries the SiPM `EFFICIENCY` as a function of photon energy, i.e. the photon detection efficiency (PDE) of an ASD-NUV4S-P-40 device, peaking at ≈ 0.43 near 420 nm, with `REFLECTIVITY` $= 0$.

The mechanism is worth stating precisely because it is the reason the sensitive detector needs no random draw of its own. In `DielectricMetal()` (lines 724–817), when the Fresnel–PDE roll places a photon in the "absorbed" channel, `DoAbsorption()` is called. Inside it (header, lines 305–320):

$$f_{\text{status}} = \begin{cases} \text{Absorption}, & \xi \geq \text{EFFICIENCY}(\lambda), \\ \text{Detection}, & \xi < \text{EFFICIENCY}(\lambda), \end{cases} \qquad \xi \sim U(0,1), \tag{1}$$

where EFFICIENCY(λ) is the tabulated PDE. The absorbed photon is assigned a non-zero `LocalEnergyDeposit` *only* when the PDE draw succeeds. Back in `PostStepDoIt`, if the status is `Detection` *and* the invoke-SD flag is set, the sensitive detector is invoked (lines 559–560):

$$f_{\text{status}} = \text{Detection} \ \wedge \ f_{\text{InvokeSD}} \implies \text{InvokeSD(step)}. \tag{2}$$

The flag is enabled in the physics list via `G4OpticalParameters::SetBoundaryInvokeSD(true)`. Consequently the sensitive-detector `ProcessHits` fires *only* for genuine photoelectrons (it tests that the step energy deposit is positive), counts one N_{pe} per call, and kills the photon (the SiPM is opaque). This is a clean separation of concerns: the physics of the photocathode lives in the optical surface, and the sensitive detector only books hits.

3.5 Per-event optical accounting

A thread-local diagnostic record (`OpticalDiag`) accumulates, per event: scintillation photons, Čerenkov photons, terminated photons, detected photoelectrons, and photons that escaped to the world. The primary muon's first-entry position in the slab is captured on the first step inside the scintillator logical volume and stored in slab-local coordinates. At end of event the sensitive detector writes one row per (event, SiPM), *including zero-detection rows*, so the sample is unbiased in the muon interaction position. Each row carries the entry position, the event's produced-photon count (the efficiency denominator), the detected count in that SiPM, the first arrival time, the mean and RMS detected wavelength, and the track weight. This one tree supports per-SiPM, combined, and spatially-binned collection-efficiency analyses.

Figure 4 shows a head-less event display of a single vertical muon in the glossy-wrapped plate (config 1). For legibility the display uses a 30 MeV muon—just below the Čerenkov threshold of the scintillator—with the scintillation yield rescaled, so that the individual optical tracks can be followed: photons are generated along the muon path, propagate by total internal reflection and specular reflection off the wrapper, and terminate on the SiPM patches at the four edges.

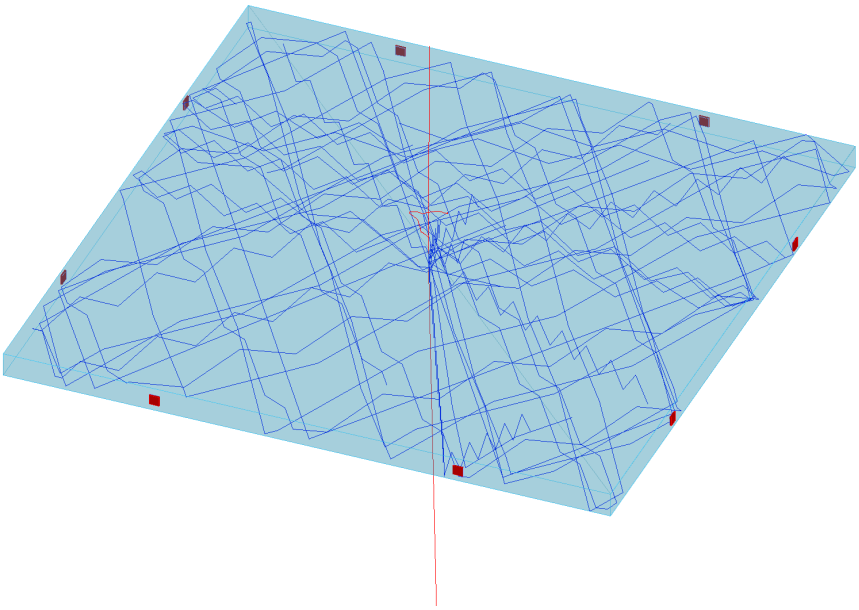


Figure 4: Head-less (`TSG_OFFSCREEN`) event display of one vertical muon in the glossy-wrapped plate with the config-1 layout, on a white background. The red line is the muon and the blue lines are optical photons (here purely scintillation, since the muon is below the Čerenkov threshold); the plate and its reflective wrapper are drawn as translucent surfaces with their edges outlined, and the SiPMs are opaque red. Both the muon energy (30 MeV, below the Čerenkov threshold) and the photon yield are chosen for visualisation only, so that individual tracks are resolved rather than merging into a saturated blob. The display is generated reproducibly by `macros/muon_vis.mac`.

4 Simulation setup and matrix scan

The primary source is a vertical muon pencil beam: a 50 mm×50 mm square beam of 4 GeV μ^- normally incident on the plate, with the beam centre placed at three positions, each irradiating a 50 mm×50 mm patch lying *entirely on* the polished plate. With the plate spanning $[-125, 125]$ mm in both x and y, the patch centres are

$$\text{centre} = (0, 0), \qquad \text{edge} = (0, -100), \qquad \text{corner} = (100, -100)\ \text{mm},$$

so the illuminated regions are $[-25, 25]^2$, $[-25, 25] \times [-125, -75]$ and $[75, 125] \times [-125, -75]$ respectively (report frame: $+y$ is the top, $-x$ the left). This places the edge and corner regions on the bottom ($-y$) edge, opposite the single instrumented face of config 3 (top, $+y$), as in Fig. 3 of Ref. [1]. It also matches the 50 mm×50 mm active area of the muon coincidence detector (MCD) that defines the regions of interest in Ref. [1], and keeps the irradiated area—and hence the denominator of Eq. 3—equal at all three positions. Each (config, reflector, position) runs 1000 events after a single `/run/initialize`. The matrix therefore comprises

$$4\ \text{configs} \times 4\ \text{reflectors} \times 3\ \text{positions} = 48\ \text{files}, \qquad 48 \times 1000 = 48\,000\ \text{events}.$$

Runs used 8 threads on an 8-thread (4-physical-core) workstation. The optional per-slab optical track-length flux map is disabled for the matrix (it is retained as a separate diagnostic); the collection efficiency is reconstructed entirely offline from the `sipmHits` tree at any desired binning, so the map is not needed.

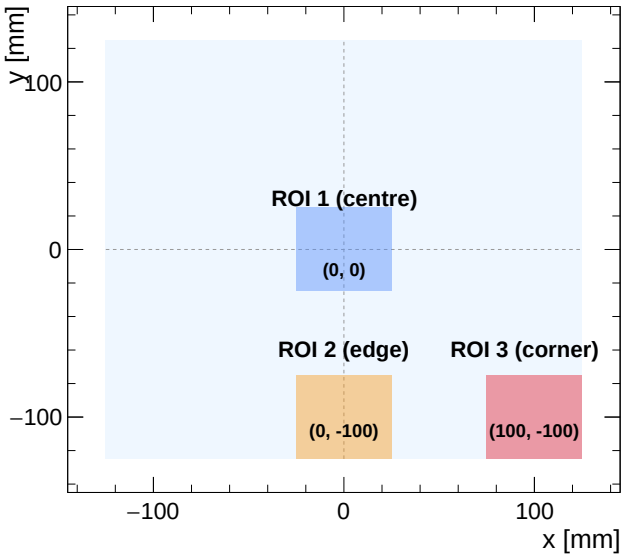


Figure 5: Plan view of the three irradiation sites (regions of interest) on the 250 mm×250 mm plate, in the report frame ($+y$ top, $-x$ left). Each coloured square is the 50 mm×50 mm footprint of the vertical muon beam: ROI 1 (centre, blue) at $(0, 0)$, ROI 2 (edge, orange) at $(0, -100)$, and ROI 3 (corner, red) at $(100, -100)$. The edge and corner sites lie on the bottom ($-y$) edge, opposite the single instrumented face of config 3.

The collection efficiency is defined, for a set $\mathcal{S}$ of SiPMs and an event e, as

$$\varepsilon = \frac{\sum_{e} \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_{e} N_{\gamma}^{(e)}}, \tag{3}$$

where $N_{\gamma}^{(e)}$ is the total number of optical photons generated in the slab during event e (scintillation plus Čerenkov) and $N_{\mathrm{pe}}^{(e,s)}$ is the number of photoelectrons in SiPM s. The all-SiPM sum over the three beam positions is quoted as the figure of merit.

For presentation, all position-resolved results (the maps of Figs. 5 and 14–17 and the whole-slab maps of Sec. 6) are shown in the report frame with the top edge at $+y$, the bottom at $-y$, the left at $-x$ and the right at $+x$. The internal simulation frame is this view rotated by 90° clockwise; the primary muon entry points and the SiPM markers are mapped into the report frame by $(x, y) \to (y, -x)$ before plotting. This rotation is a rigid change of view and leaves every scalar result (yields, spectra, efficiencies) unchanged; it only fixes the orientation of config 3 (top edge) and of the edge/corner irradiation sites (bottom edge).

5 Results

5.1 Overall collection efficiency

Figure 6 shows the overall collection efficiency (Eq. 3, all SiPMs, all three beam positions pooled) for every (config, reflector) pair, and Table 2 lists the numbers. The ordering

$$\text{bare}\ (6{\times}10^{-6}\text{–}2{\times}10^{-4}) \ \ll\ \mathrm{TiO_2}\ (2{\times}10^{-4}\text{–}10^{-2}) \ <\ \mathrm{Al}\ (1.4\text{–}2.6{\times}10^{-2}) \ \approx\ \text{glossy}\ (1.5\text{–}3.0{\times}10^{-2})$$

holds for all four layouts, reproducing the experimental result of Ref. [1]. The bare configuration is suppressed by two to three orders of magnitude, as required by the frustrated-TIR absorber

model: without a reflective wrap, the trapped light is lost to the black holder. Al and glossy paper are statistically indistinguishable within the 1000-event-per-position sample and are the best collectors. Four SiPMs on four faces (config 2) outperform two SiPMs on one face (config 3), and the eight-SiPM layout (config 1) gives the highest efficiency, as expected from the larger photosensitive coverage. The single-SiPM WLS-fibre layout (config 4) collects substantially less than the direct-coupled layouts: only the fraction of light that reaches the fibre face is wavelength-shifted and guided to the sensor, so its efficiency sits about an order of magnitude below config 3.

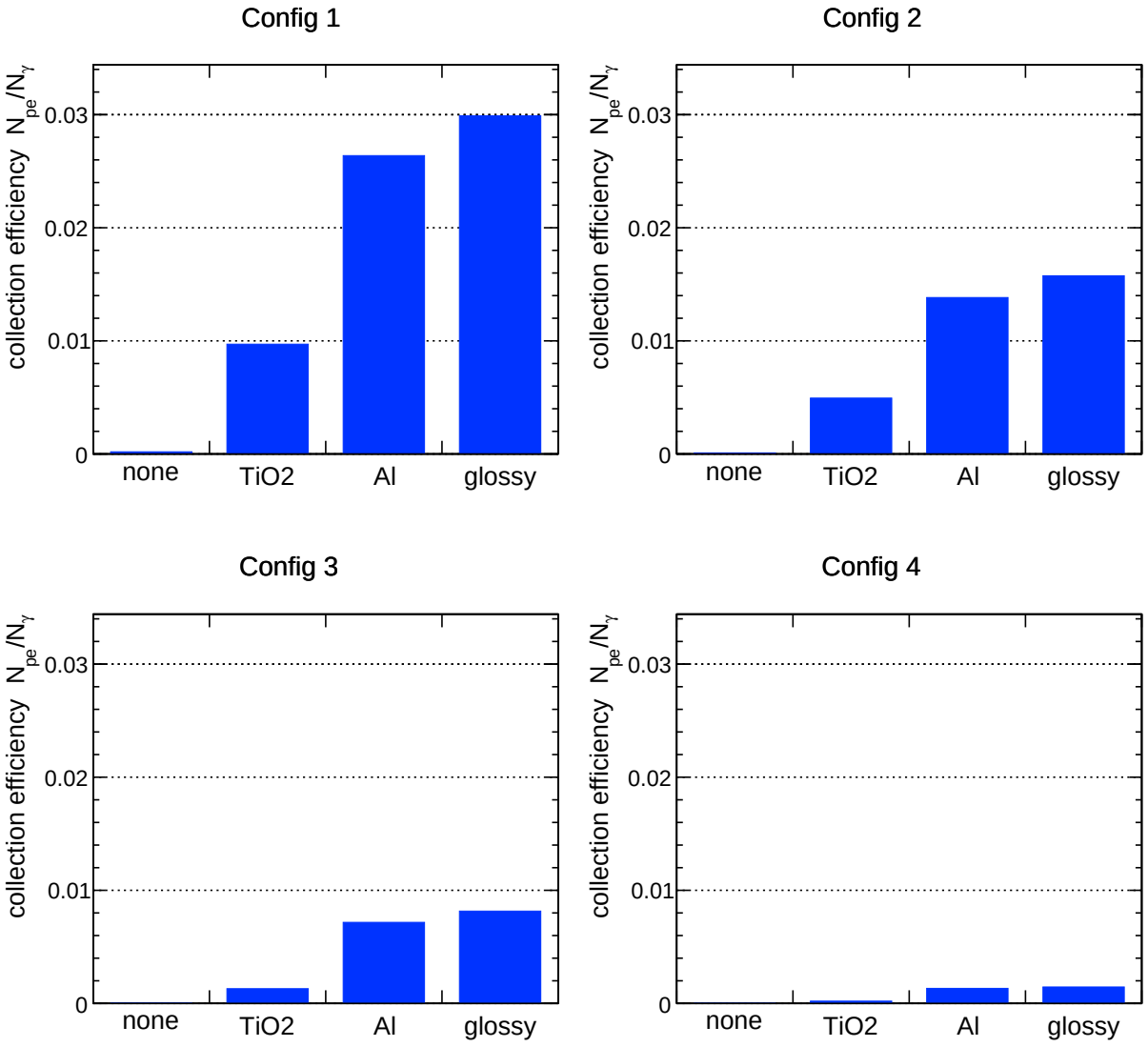


Figure 6: Overall collection efficiency $\varepsilon = \sum N_{\mathrm{pe}} / \sum N_\gamma$ for the four reflector treatments and four detector layouts. The ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy is reproduced in every layout. Error bars assume Poisson counting of photoelectrons.

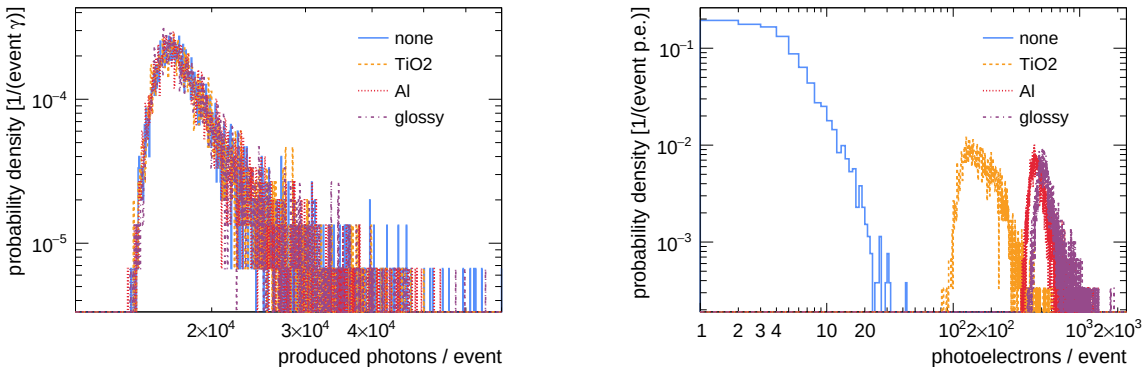
Table 2: Overall collection efficiency (all SiPMs, all beam positions pooled). All beam patches lie fully on the plate.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	1.9×10^{-4}	9.7×10^{-3}	2.64×10^{-2}	2.99×10^{-2}
Config 2 (4 SiPMs)	9.7×10^{-5}	4.9×10^{-3}	1.38×10^{-2}	1.58×10^{-2}
Config 3 (2 SiPMs)	2.3×10^{-5}	1.30×10^{-3}	7.17×10^{-3}	8.16×10^{-3}
Config 4 (1 SiPM, WLS)	6.4×10^{-6}	2.05×10^{-4}	1.33×10^{-3}	1.46×10^{-3}

5.2 Photon yield and photoelectron spectra

Figure 7a shows the per-event optical-photon yield (the denominator of Eq. 3) for the four reflectors. The mean yield is essentially independent of the reflector—about 19 000 photons per event—which is the correct physical behaviour: the reflector changes how photons are transported, not how many are produced. The small spread reflects event-to-event variation in the muon path length.

Figure 7b shows the corresponding photoelectron spectra. Here the reflector has a dramatic effect: the bare configuration peaks near zero with a mean of only 3.8 N_{pe}/event, whereas glossy paper has a mean of 574 N_{pe}/event. The shift from a near-zero-peaked to a broad distribution is the direct signature of effective light collection.



(a) Yield (produced photons/event). **(b)** Photoelectrons/event.

Figure 7: Per-event distributions for config 1, plotted as a probability density (unit area, 1 event per unit of the abscissa) with logarithmic x- and y-axes. The production yield (left) is reflector-independent, as it must be; the detected photoelectron count (right) depends strongly on the wrapper. The log y-axis is needed because the four curves have the same event count but very different widths: the concentrated bare peak and the broad reflective distributions cannot both be resolved on a linear scale.

The efficiency scale is set by the SiPM PDE and the coupling: of order 19 000 photons are produced and the best wrapper delivers ≈ 574 photoelectrons, i.e. $\approx 3\%$. Given a PDE of ~ 0.4 and the geometric acceptance of a side-mounted sensor, this is consistent.

5.3 Wavelength of detected photons

Figure 8 shows the wavelength distribution of detected photoelectrons, weighted by their count, for the four reflectors. The distribution peaks near 460 nm for the reflective wrappers (measured mean 459.5 ± 4.7 nm for Al, config 1), i.e. slightly red-shifted with respect to the 425 nm emission peak. This is the expected consequence of self-absorption: the shorter-wavelength part of the emission overlaps the absorption band and is attenuated over the multi-bounce path, so the surviving and detected spectrum is red-shifted. The bare configuration has too few detected photons for a meaningful spectrum.

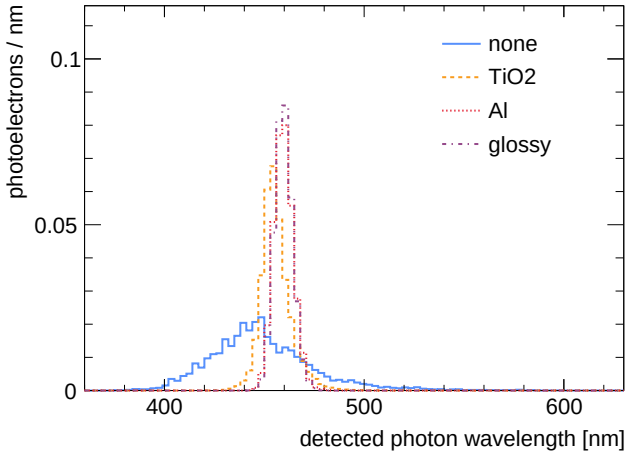


Figure 8: Wavelength distribution of detected photoelectrons (config 1), weighted by detected count. The reflective wrappers give a distribution peaking near 460 nm, red-shifted from the 425 nm emission peak by self-absorption over the multi-bounce path.

Figure 9 compares the detected-photon spectrum of the four configurations for each wrapper. Configurations 1–3, in which the SiPMs view the tile directly, all detect the same blue scintillation spectrum: the *mean* wavelength is essentially independent of the number of sensors, moving only from 459.5 nm (glossy, config 1) to 460.3 nm (config 3), the small increase reflecting the slightly longer collection path of the single-sided layouts. Config 4 is qualitatively different: the WLS fibres absorb the blue light and re-emit it at their own emission band, so the detected spectrum is shifted to 480.6 nm (glossy) — a green-shift of some 20 nm. The same shift appears for every wrapper, including the bare tile (which, with its very small photon count, is statistically noisy), confirming that it originates in the fibre wavelength shifter rather than in the wrapper.

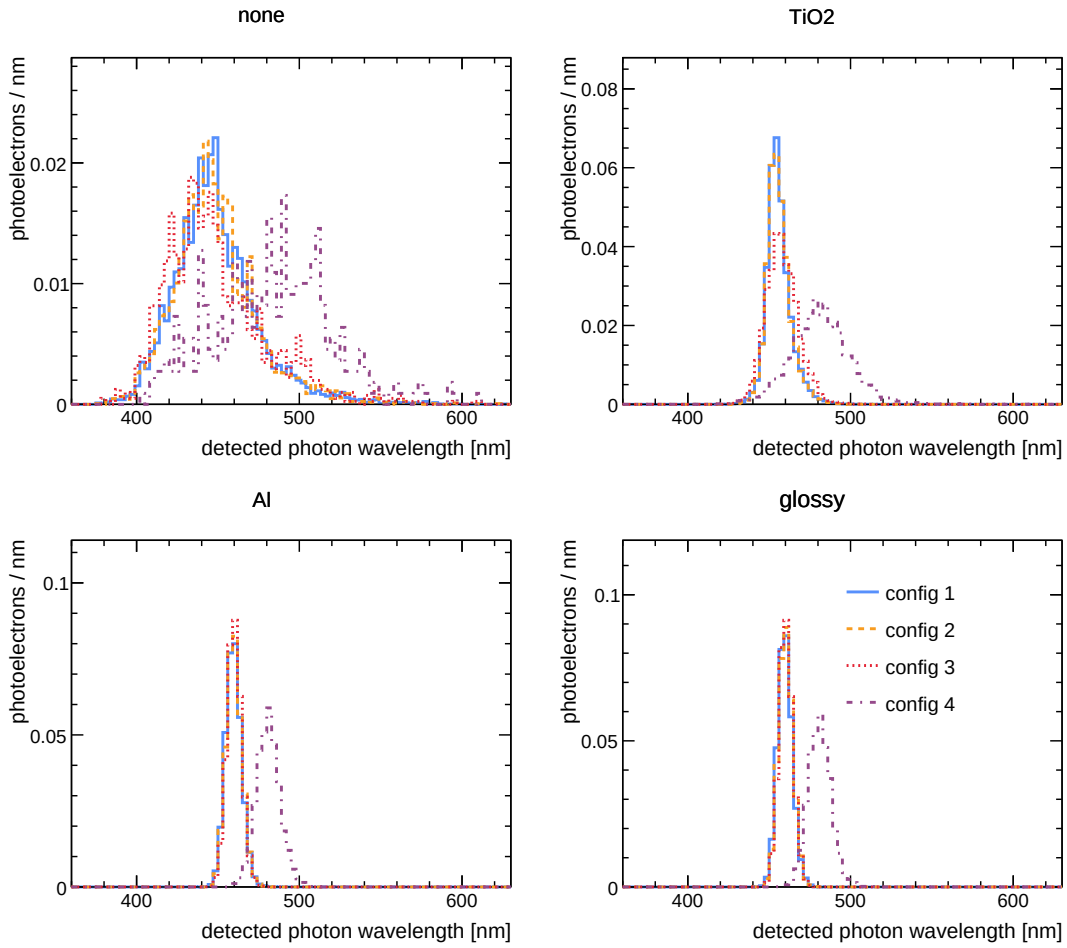


Figure 9: Detected-photon wavelength distribution, weighted by photoelectron count, for each wrapper (panel) and each configuration (curves), using the pooled data of the three irradiation positions. Configurations 1–3 follow the 460 nm blue scintillation spectrum regardless of sensor count, whereas the WLS-fibre readout of config 4 is green-shifted to ≈ 480 nm by the Y-11(200) emission band.

5.4 Spatial and directional dependence

The matrix scan separates the three 50 mm×50 mm patches (centre, edge, corner) precisely so that the response can be resolved by position. The tempting single-coordinate summary is the distance r of the patch from the plate centre, but r is blind to azimuth, and the shortcoming is decisive here: configs 3 and 4 instrument a *single* face, so the collection depends on the *direction* from which the plate is irradiated, not merely on the distance. All three patches also lie on the half of the plate opposite the instrumented face of the single-sided layouts, so their distances are not even monotonic in the proximity to a sensor. We therefore keep the three patches separate and compare them directly.

Figure 10 groups the three patches by layout (one panel per config, four reflector curves) and Fig. 11 groups them by reflector (one panel per wrapper, four layout curves). Because every patch lies fully on the plate and all three have equal area, the produced-photon denominators are equal and the three points within each curve are directly comparable. The curves are plotted on a logarithmic y-axis, so the weakly collecting bare plate is resolved alongside the reflective wrappers.

The conspicuous feature is that the reflective wrappers *homogenise* the response: for Al and glossy paper the efficiency varies by at most 1 % to 9 % across the three patches in every layout (Fig. 12 and Table 3), i.e. an edge- or corner-irradiated plate collects almost exactly as much light

as a centre-irradiated one. To quantify this we define the *position contrast*

$$\kappa \equiv \frac{\varepsilon_{\max}}{\varepsilon_{\min}}, \tag{4}$$

the ratio of the largest to the smallest of the three patch-averaged collection efficiencies ε (Eq. 3), so that $\kappa = 1$ denotes a perfectly uniform response and $\kappa - 1$ the fractional spread. The diffuse TiO_2 paint is markedly less uniform (a factor ≈ 1.7–1.9), and the bare plate records by far the strongest position dependence (up to a factor 8 for config 4, whose single readout amplifies any asymmetry). This is the expected behaviour of a closed reflecting cavity: after the first few bounces the light distribution no longer remembers where the muon crossed the plate, so the sensor coverage, and not the irradiation point, sets the collection. A single-sided layout is the extreme test of this argument, and even there the mirror wrappers keep the position contrast below 10 %.

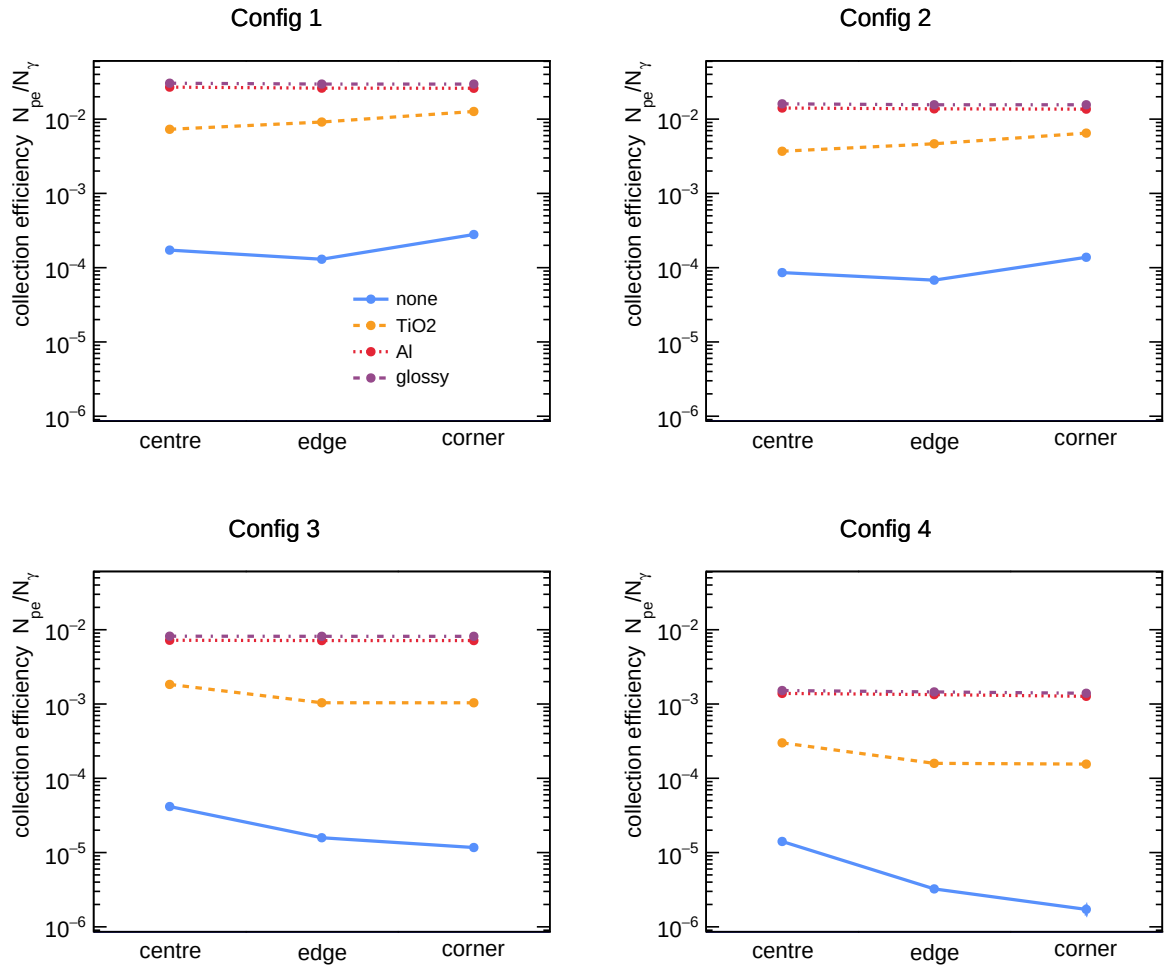


Figure 10: Collection efficiency at the three irradiation patches (centre, edge, corner), one panel per detector layout, with the four reflector curves (logarithmic y-axis). All three patches have equal area and lie fully on the plate, so the points are directly comparable. The reflective wrappers are nearly flat across the three patches in every layout; the bare plate is both weakest and most position-dependent.

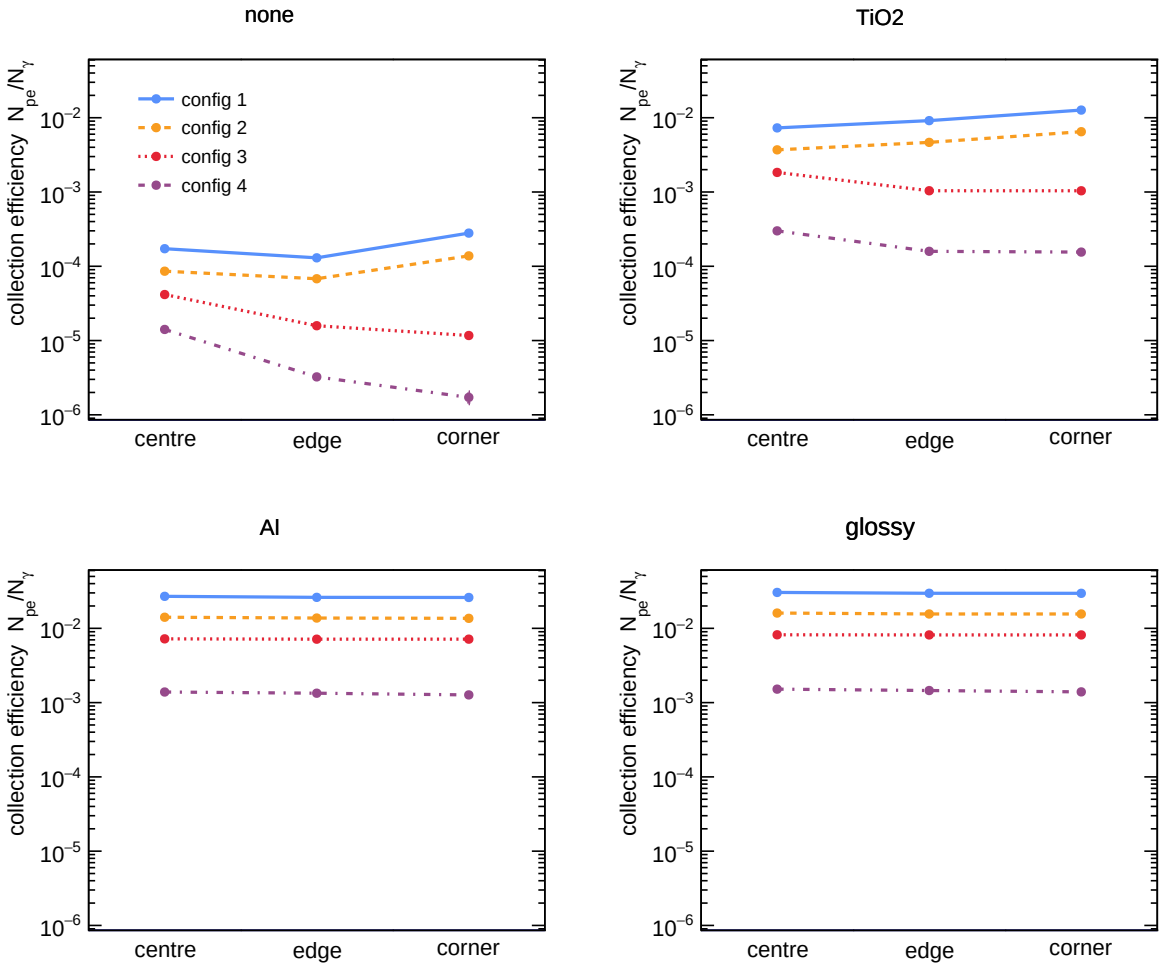


Figure 11: The same data grouped by reflector (one panel per wrapper), with the four layouts as curves. Configs 1 and 2 (sensors on every face) are the most uniform; the single-sided layouts (3 and 4) drop toward the edge and corner patches, which sit on the far side of the plate from their sensors.

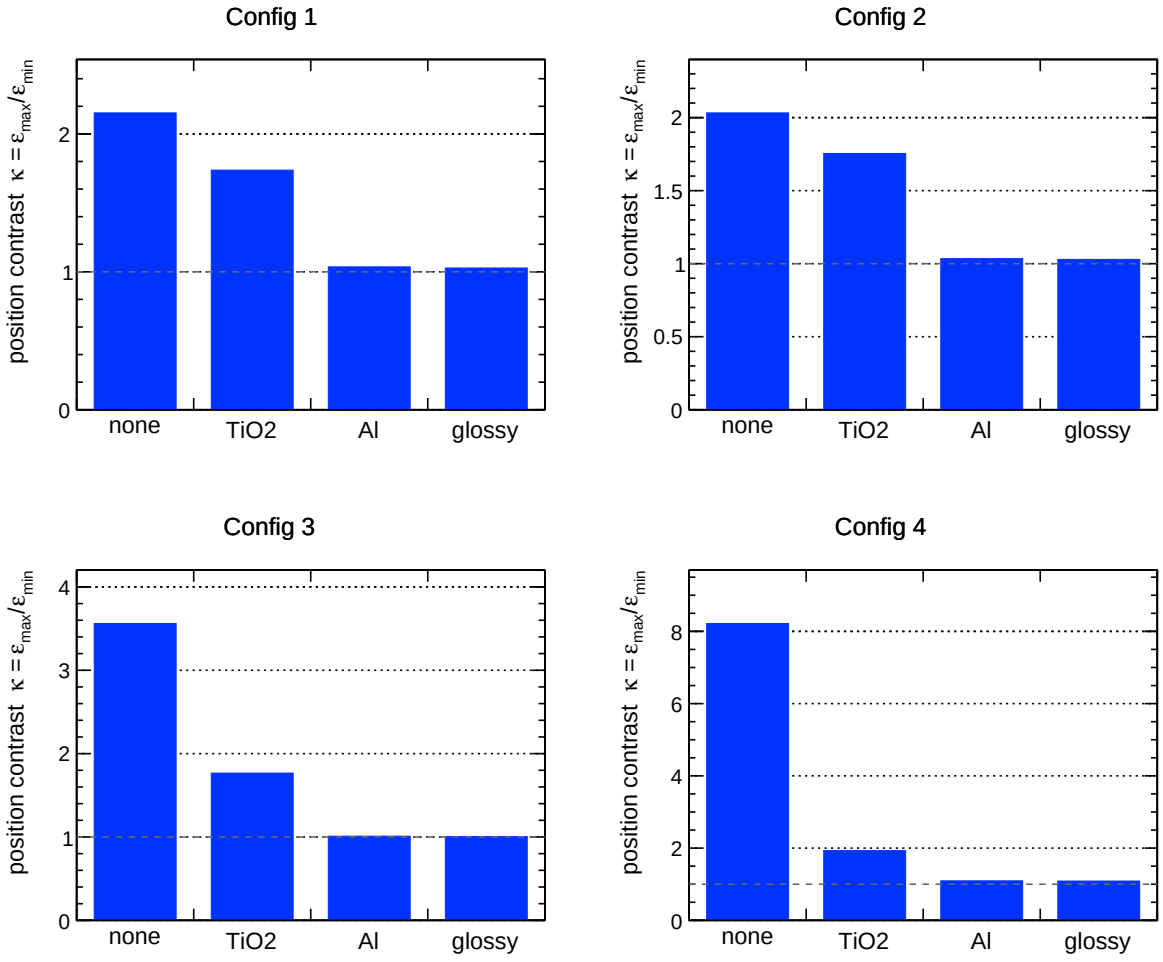


Figure 12: Position contrast $\kappa = \varepsilon_{\max}/\varepsilon_{\min}$ (Eq. 4) over the three irradiation patches, one panel per layout, four reflector bars (dashed line: $\kappa = 1$). The mirror-like wrappers hold the contrast within 9 % in every layout, whereas diffuse TiO_2 and especially the bare plate are strongly position-dependent.

Table 3: Position contrast $\kappa = \varepsilon_{\max}/\varepsilon_{\min}$ (Eq. 4) across the three irradiation patches (centre, edge, corner). A value of 1 means the layout collects the same fraction of light wherever the muon crosses the plate.

Reflector	Config 1	Config 2	Config 3	Config 4
Bare	2.15	2.03	3.56	8.22
TiO_2	1.74	1.75	1.77	1.93
Al	1.04	1.03	1.01	1.09
Glossy	1.03	1.03	1.00	1.09

Figure 13 resolves the directional response most directly. For config 1 (the layout with the most sensors) it shows the *share* of the collected light seen by each of the eight sensors, in one panel per reflector, with the three irradiation patches as grouped bars (dashed line: the uniform expectation of 1/8). The difference between the wrappers is stark. For the *bare* plate the light barely mixes: at the corner patch the two sensors nearest the beam (B2, R1) together collect 91 % of all photoelectrons, i.e. some 3.6 times the uniform share each, while the remaining six share only 9 % and two of them (B1, R2) record nothing at all. A diffuse or mirror wrapper erases this structure: for glossy paper the same two nearest sensors reach only about 1.1 times the uniform share. The total efficiency is nearly the same for all three patches (Table 3), so

the cavity keeps the *amount* of collected light constant while completely rearranging its spatial division. A radius-only summary averages these distinct per-sensor patterns into a single number and hides precisely this information.

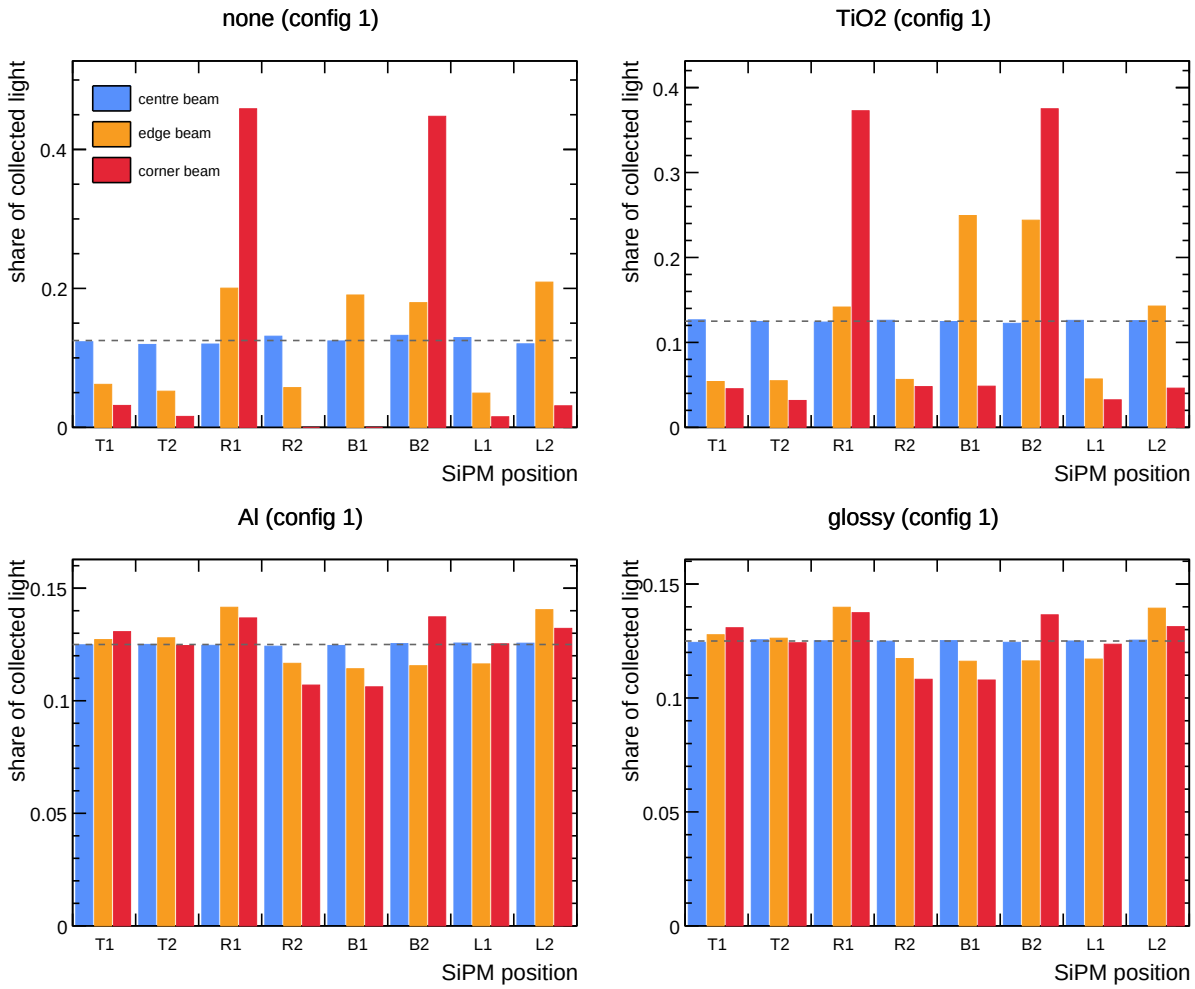


Figure 13: Direction-aware light sharing, config 1, one panel per reflector. Each group of bars is one sensor (x axis) and the three colours are the irradiation patches (blue: centre, orange: edge, red: corner); the height is that sensor's share of the light collected at that patch, and the dashed line marks the uniform expectation 1/8. For the bare plate the light stays concentrated on the two sensors nearest the beam, whereas the reflective wrappers spread it almost uniformly over all eight.

5.5 Position-resolved efficiency maps

Figures 14–17 show the two-dimensional collection efficiency as a function of the muon entry point (x, y), one panel per reflector, for the four configurations. Each map is the ratio of collected photoelectrons to produced photons in $5 \times 5\,\mathrm{mm}$ cells (Eq. 3, evaluated bin-by-bin). The black squares mark the SiPM positions for that configuration (same convention as Fig. 1). Because the matrix scan only irradiates the three $50\,\mathrm{mm} \times 50\,\mathrm{mm}$ patches, only those regions are populated; the maps therefore show the internal structure of each of the three fields. Within a patch the reflective wrappers are almost featureless—consistent with the near-uniform integrated response found above—while the bare plate and, to a lesser degree, the TiO_2 coating show a clear gradient across the field, brightest on the side nearest the sensors.

Table 4 lists the absolute centre/edge/corner efficiencies for config 1; the position- and direction-resolved comparison across all four layouts is the subject of the preceding subsection (Figs. 10–13), and the TiO_2 corner excess it reveals is discussed in Sec. 7.

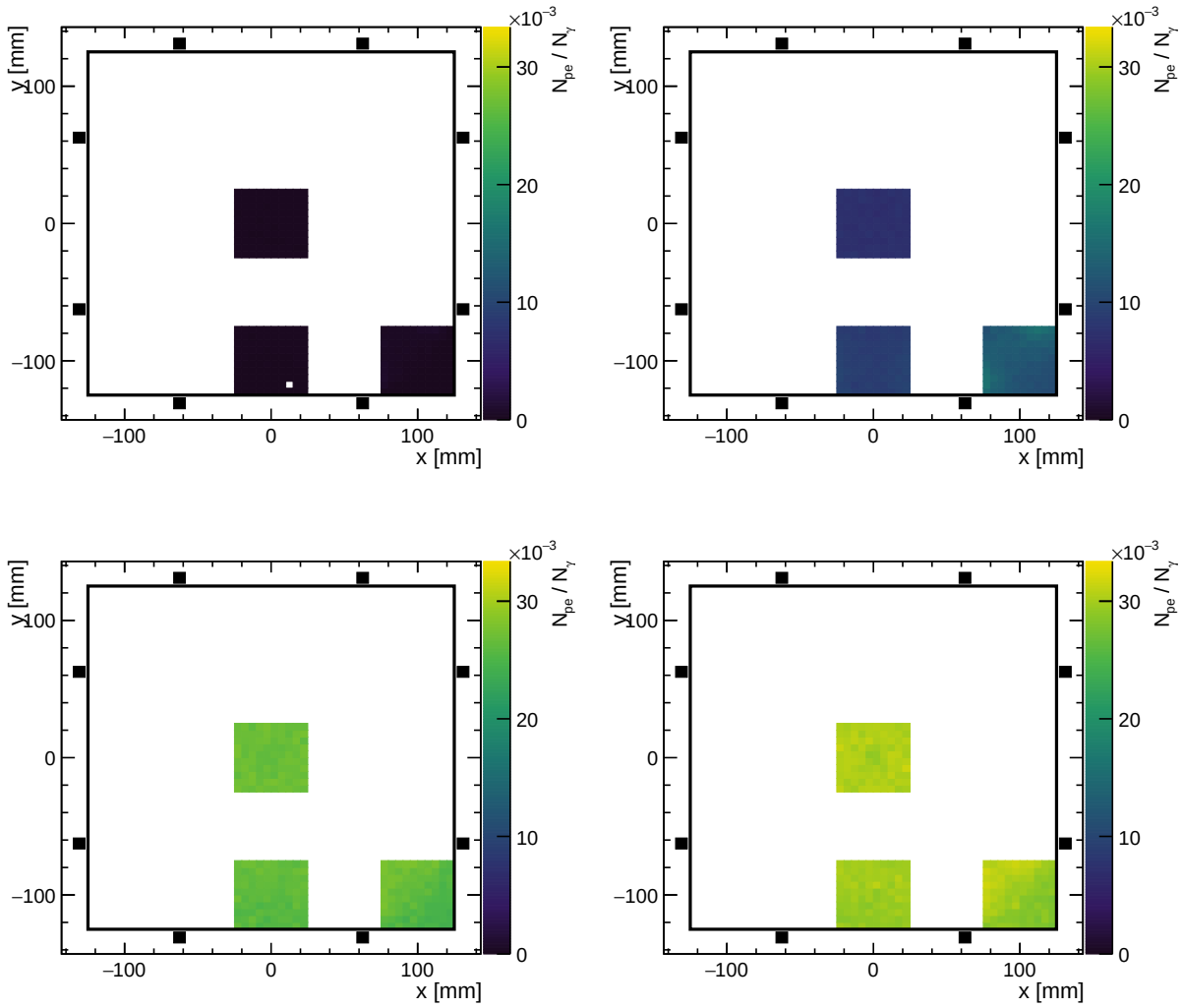


Figure 14: Collection efficiency map, config 1 (8 SiPMs), one panel per reflector (bare, TiO_2, Al, glossy). Bins are 5×5 mm; black squares mark the SiPM positions (two per edge).

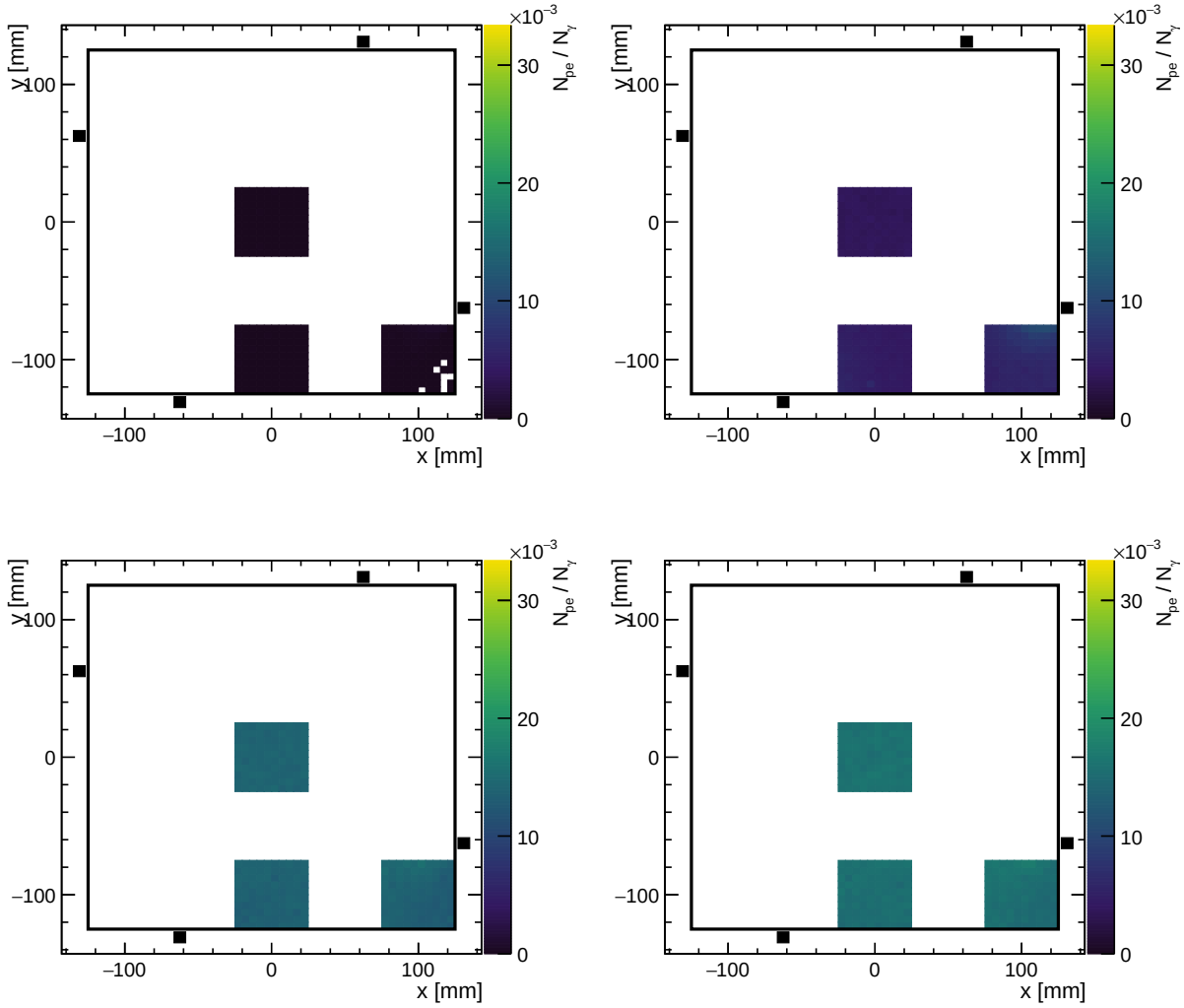


Figure 15: Collection efficiency map, config 2 (4 SiPMs, one per side). Black squares mark the SiPM positions.

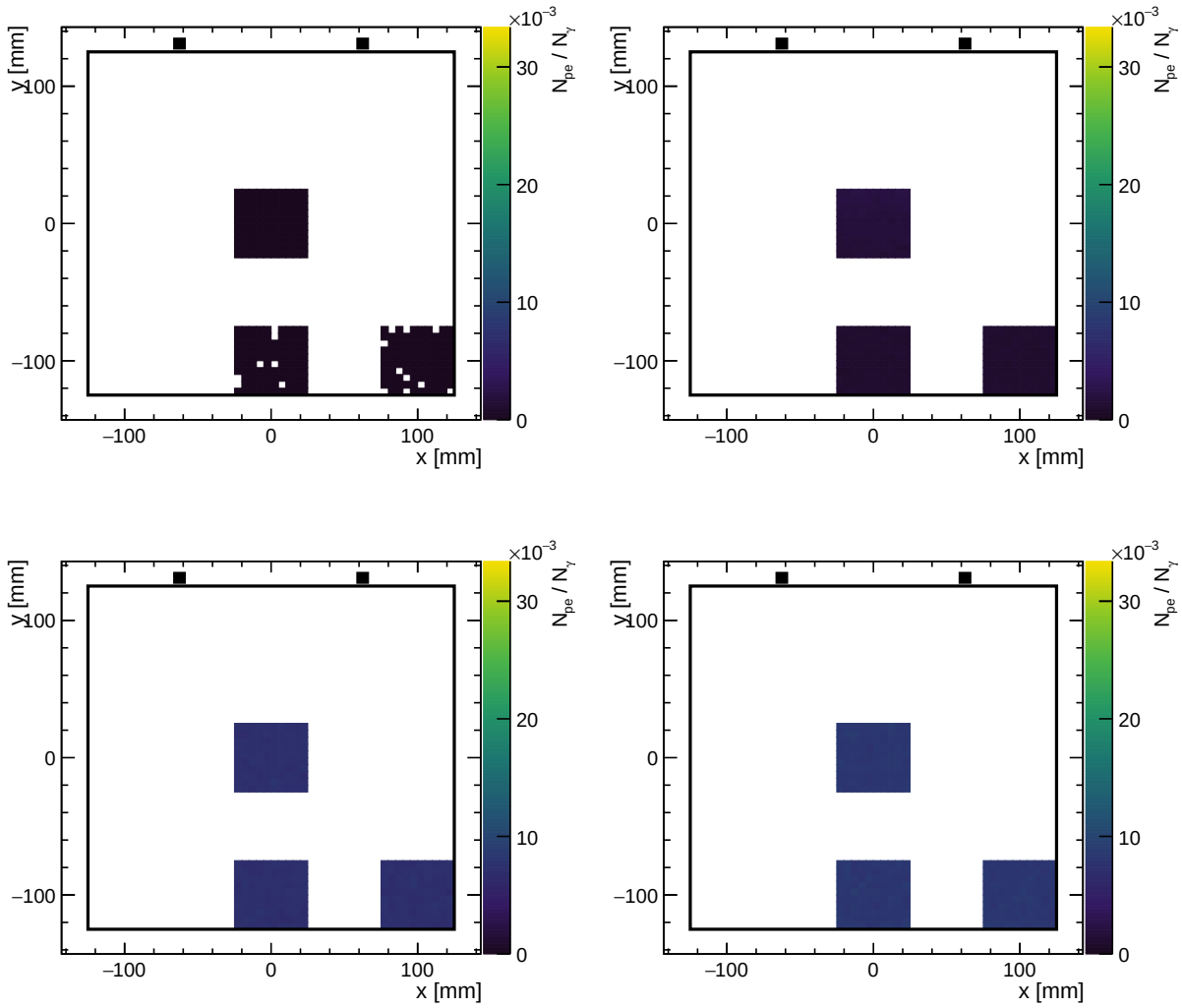


Figure 16: Collection efficiency map, config 3 (2 SiPMs, both on the top $+y$ side). Black squares mark the SiPM positions.

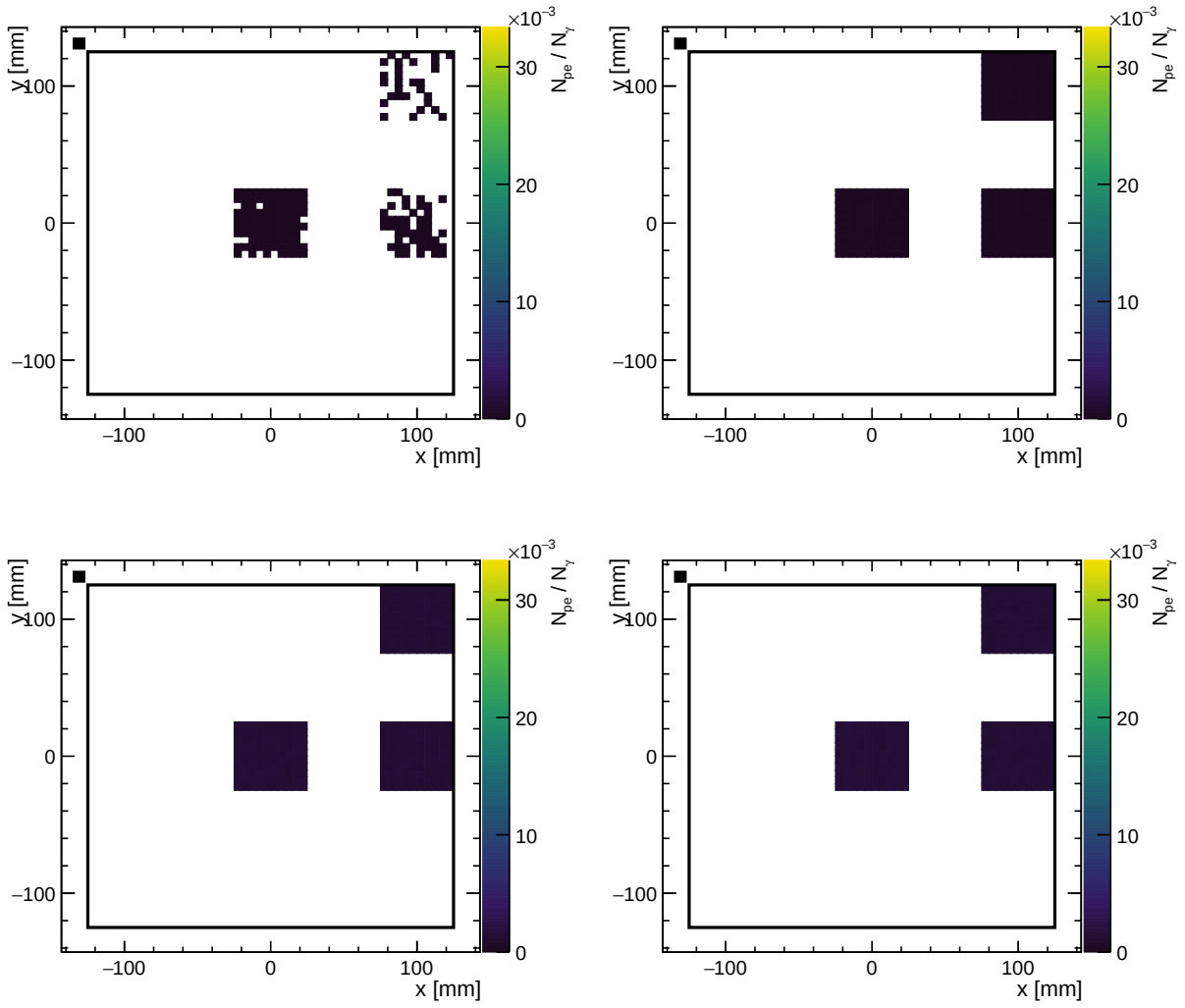


Figure 17: Collection efficiency map, config 4 (single SiPM + WLS fibre clamp). The black square marks the read-out SiPM position. (Config 4 is still under development; this panel uses the internal simulation frame and is shown for completeness only.)

Table 4: Position dependence of the collection efficiency, config 1 (all SiPMs). Each beam patch is 50 mm×50 mm and lies fully on the plate, so the produced-photon denominators are equal.

Reflector	Centre	Edge	Corner
Bare	1.7×10^{-4}	1.3×10^{-4}	2.8×10^{-4}
TiO_2	7.3×10^{-3}	9.1×10^{-3}	1.27×10^{-2}
Al	2.70×10^{-2}	2.61×10^{-2}	2.60×10^{-2}
Glossy	3.04×10^{-2}	2.97×10^{-2}	2.96×10^{-2}

6 Whole-slab irradiation

The matrix scan illuminates only three 50 mm×50 mm patches, so it resolves the light collection at the centre, edge and corner but not the full plate. To obtain a complete position-resolved map we ran a second, *whole-slab* configuration: the GPS plane is expanded to 250 × 250 mm (`halfx` = `halfy` = 125 mm), so the primary muon's first-entry position is sampled *uniformly* over the entire plate. Ten thousand events are generated per (config, reflector); the efficiency is reconstructed

from the same `sipmHits` tree, in 10×10 mm cells, as

$$\varepsilon(x, y) = \frac{\sum_{e \in \text{cell}} N_{\text{pe}}^{(e)}}{\sum_{e \in \text{cell}} N_{\gamma}^{(e)}},$$

i.e. identically to Eq. 3 but binned in the entry position. Because the beam now covers the whole plate, the whole-slab maps fill in the gaps left by the three-patch scan.

Figure 18 shows the complete set of full-plate maps for config 1, one panel per reflector. For the bare plate, averaged over the whole surface the efficiency is $\varepsilon = 4.7 \times 10^{-4}$, consistent with the three-patch bare values of Table 4. The map reveals a clear radial structure that the three discrete patches could only hint at: the efficiency is lowest at the centre ($\varepsilon \approx 1.8 \times 10^{-4}$ within $r < 60$ mm) and rises monotonically toward the edges ($\varepsilon \approx 5.4 \times 10^{-4}$ in the outer region), i.e. by roughly a factor of three. This is exactly the signature of edge readout combined with frustrated-TIR absorption: photons created near a sensor-bearing face have a short path to detection, whereas photons born at the centre must random-walk to an edge through the lossy holder contact, and most are absorbed before arriving.

The reflective panels show the same ordering as the matrix scan, with the position dependence progressively erased. TiO_2 is about twenty times brighter than the bare plate ($\varepsilon = 1.08 \times 10^{-2}$) and its centre-to-edge gradient is much weaker (7.7×10^{-3} for $r < 60$ mm versus 1.15×10^{-2} beyond, a factor of $\sim$1.5). Al is brighter again (2.69×10^{-2}) and almost perfectly *flat*: the centre (2.67×10^{-2}) equals the outskirts (2.70×10^{-2}). Glossy is the brightest (3.04×10^{-2}) and likewise flat (3.03 vs. 3.05×10^{-2}). Diffuse recycling (TiO_2) and specular mixing (Al, glossy) both reduce the penalty of being far from a sensor, the specular mirrors so thoroughly that the entry position is effectively erased, whereas the bare plate—whose frustrated TIR leaks into the holder—retains a steep edge gradient.

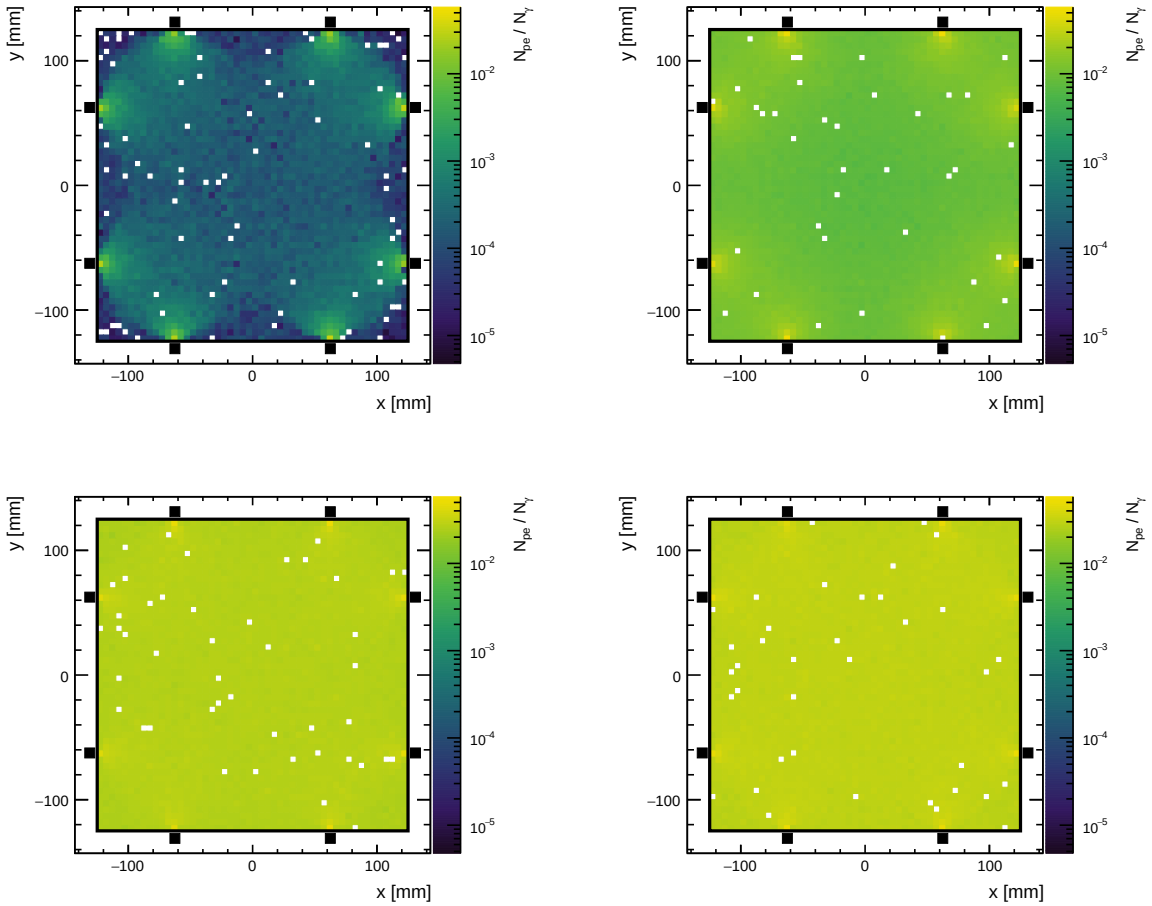


Figure 18: Full-plate collection-efficiency maps for config 1, from 10 000 whole-slab events per panel (10×10 mm cells, on a logarithmic colour scale shared by the panels). Bare plate (top left) rises from the centre toward the instrumented edges; TiO_2 (top right) is roughly twenty times brighter with only a weak gradient; Al (bottom left) and glossy (bottom right) are brighter still and almost flat. The bold rectangle is the 250 mm×250 mm slab outline and the black squares mark the eight SiPM positions.

Figure 19 repeats the whole-slab scan for config 2 (four sensors, one on each lateral face at the config-1 offsets). Every configuration is weaker than its config-1 counterpart—roughly halved, as expected from the reduced sensor area—while the reflector ordering and the flattening behaviour are unchanged. The bare plate gives $\varepsilon = 2.6 \times 10^{-4}$ with the same strong edge gradient (1.3×10^{-4} at the centre versus 2.9×10^{-4} outside $r = 60$ mm); TiO_2 reaches 5.55×10^{-3} with a milder gradient, and Al and glossy reach 1.42×10^{-2} and 1.62×10^{-2}, both nearly flat. The monotone bare $\ll$ TiO_2 $<$ Al $\approx$ glossy sequence and the progressive erasure of the position dependence are therefore robust against the sensor count: fewer sensors lower the overall light collection but do not change the physics of the wrapper.

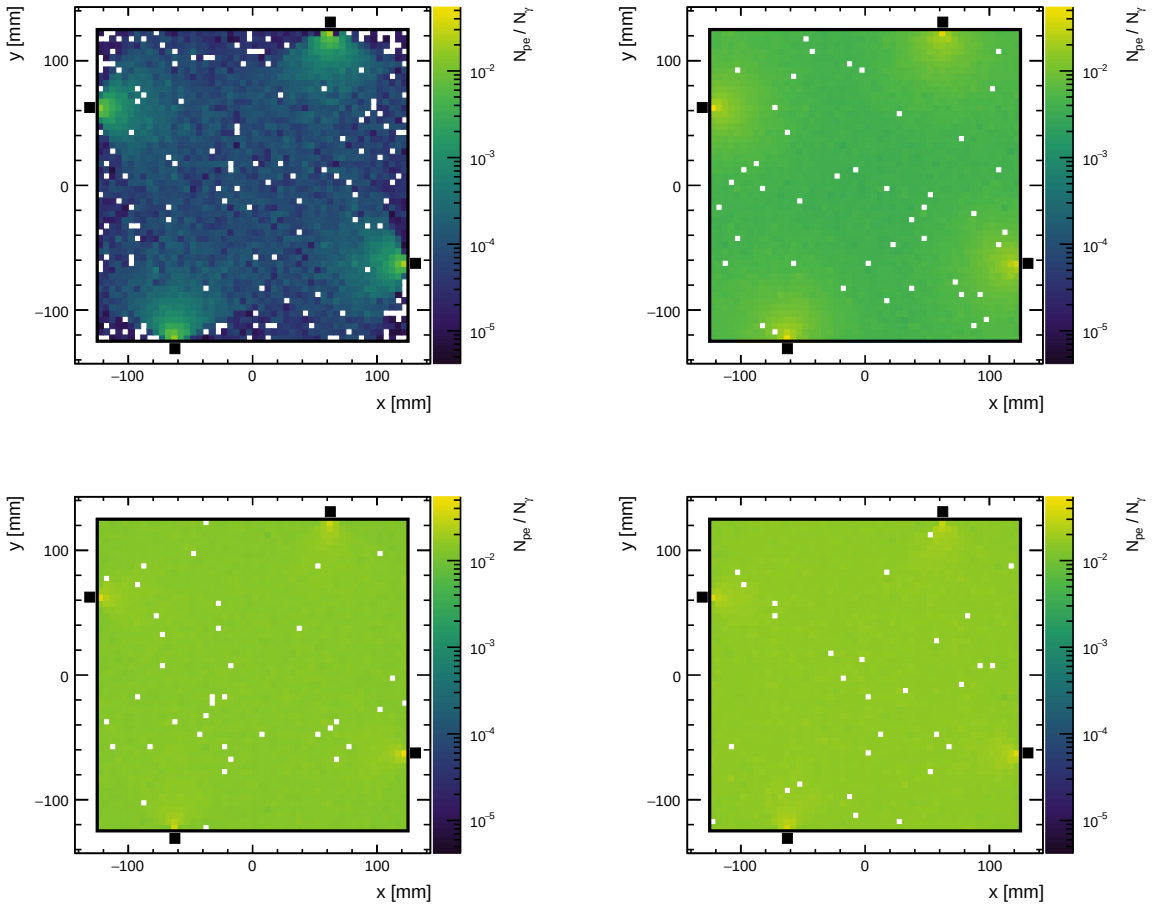


Figure 19: Full-plate collection-efficiency maps for config 2 (four SiPMs, one per lateral face at the config-1 offsets), otherwise identical to Fig. 18. The reflector ordering and the centre-to-edge flattening are the same, with each efficiency about half the config-1 value.

Finally, Fig. 20 shows config 3, in which both sensors sit on the same (top, $+y$) face. The integrated efficiencies are the lowest of the direct-coupled configurations—bare 1.1×10^{-4}, TiO_2 2.7×10^{-3}, Al 7.2×10^{-3}, glossy 8.2×10^{-3}—because concentrating both sensors on one face halves the solid angle available to photons born on the opposite side. The maps make the resulting asymmetry visible: the bare and TiO_2 panels are brightest near the instrumented top ($+y$) edge and dimmest on the far ($-y$) side (4.4×10^{-5} at the centre versus 1.3×10^{-4} outside $r = 60\,\text{mm}$ for the bare plate). The specular wrappers again homogenise the response, with Al and glossy flat to better than 1% between the inner and outer regions.

The whole-slab irradiation was carried out for the direct-coupled layouts only (configs 1–3); the WLS-fibre layout (config 4) is still under development and has no full-plate run. Across the three direct-coupled layouts the reflector ordering is identical and monotone,

$$\text{bare} \ll \text{TiO}_2 < \text{Al} < \text{glossy},$$

and the efficiency scales roughly with the collected photosensitive area, confirming that the wrapper, not the readout layout, governs the collection efficiency.

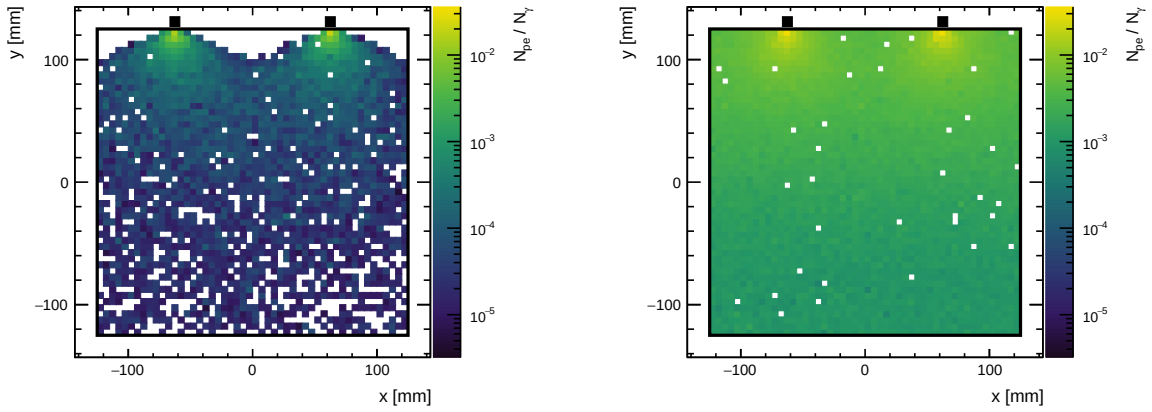


Figure 20: Full-plate collection-efficiency maps for config 3 (two SiPMs, both on the top $+y$ face). The lowest efficiencies of the direct-coupled layouts and the clearest top–bottom asymmetry appear here; the specular wrappers (Al, glossy) again flatten the response.

7 Caveats and open items

Intermittent multi-threaded stall. During the matrix production runs a rare stall was observed: on one occasion a single worker thread entered a tight loop (100% CPU) while the remaining threads waited at the event-level barrier, freezing the run. The condition is intermittent—the same macro completed normally on a clean re-run—and is consistent with a pathological optical track (e.g. a photon bouncing in a near-degenerate configuration) rather than a deterministic defect. Because a single stalled task would consume the entire wall-time allocation on a batch system, the production workflow wraps every task in a timeout with automatic retry. The matrix results reported here are unaffected: all 36 files completed, and the frozen run was discarded and regenerated.

Reversed position trend in TiO_2. For the mirror-like wrappers (Al, glossy) the response is essentially flat across the three patches (contrast $< 9\,\%$, Table 3), with the centre marginally the most efficient and the corner the least, consistent with the shortest average path to a sensor. The TiO_2 configuration instead shows a clear corner > edge > centre ordering (a factor 1.7). Since `groundfrontpainted` is purely Lambertian (no TIR, no specular reflection), the sharp corner provides two orthogonal painted walls close to the sensor-bearing edges, and diffusely scattered light is re-emitted into the plate rather than specularly returned; the corner geometry can therefore couple more light into the edge sensors. The effect persists at the higher 1000-event-per-position statistics used here, though no corresponding trend is reported in Ref. [1] and it should be confirmed with independent simulation settings before being interpreted physically.

Scale and normalisation. The absolute efficiencies are model-dependent: they are fixed by the assumed PDE curve, the grease coupling, and the absorbed-fraction of the tabulated reflectivities. The robust result is the *ordering* and the qualitative splitting into an effectively absorbing bare case and a reflective family.

8 Summary

A Geant4 optical simulation of a $250\,\text{mm} \times 250\,\text{mm} \times 10\,\text{mm}$ polished plastic scintillator read out by side-mounted SiPMs has been developed and applied to a full matrix of four SiPM layouts and four reflector treatments at three beam positions. The optical physics—unified surface model,

finish selection, photocathode PDE via the boundary process, and per-event photon accounting—has been verified against the Geant4 source. The results reproduce the experimentally observed ordering bare $\ll$ TiO_2 $<$ Al $\approx$ glossy, with the bare plate suppressed by two to three orders of magnitude by frustrated total internal reflection into a black holder. A complementary whole-slab irradiation samples the full 250 mm$\times$250 mm plate uniformly and yields a complete collection-efficiency map, which for the bare configuration shows the expected factor-of-three rise from centre to edge. Keeping the three irradiation patches separate—rather than collapsing them to a single beam-axis radius—reveals that the reflective wrappers homogenise the response: the position contrast stays below 9 % in every layout, and the per-sensor light sharing becomes nearly uniform, whereas the bare plate concentrates 91 % of the collected light on the two sensors nearest the beam.

References

[1] Rao *et al.*, J. Appl. Phys. **138**, 114504 (2025).

[2] Geant4 Collaboration, *Physics Reference Manual: Optical Processes*, Geant4 v11.3.2.

[3] Eljen Technology, *EJ-200 General Purpose Plastic Scintillator* datasheet.

[4] Amcrys, *UPS-923A Plastic Scintillator* datasheet.

[5] J. B. Birks and F. A. Craun, *The theory and practice of scintillation counting* (1970).